

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3) Assessment Tool Weightage: 10%

Code of Conduct:

I M V Veeranjneya Reddy (192472140) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M V Veeranjneya Reddy

Rubrics for Evaluation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks(100)
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	

Accuracy	10%	Completely accurate and	Mostly accurate with	Partially correct with missing	Incorrect answers with	
		<i>covers all required points</i>	<i>minor omissions</i>	<i>points</i>	<i>major gaps</i>	
Clarity	5%	<i>Clear, neat, professional presentation</i>	<i>Understandabl e with minor issues</i>	<i>Limited clarity, readability issues</i>	<i>Poorly written, difficult to understand</i>	

1.An image appears too bright.

Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process, and expected outcome.

ANS:- Most Suitable Grey Level Transformation: Image Negative (for bright backgrounds) or Log/Power-Law with Gamma > 1.

For a generally **over-bright image**, the most commonly recommended transformation is **Power-Law (Gamma Correction) with $\gamma > 1$** , as it reduces brightness while preserving details.

Concept

Grey level transformation changes the intensity values (pixel values) of an image to improve its appearance.

In **Gamma Correction**, the output pixel value is calculated as:

$$s = c \cdot r^{\gamma}$$

where:

- **r** = input pixel value (normalized between 0 and 1)
- **s** = output pixel value
- **c** = constant (usually 1)
- **γ (gamma)** = gamma value

When **$\gamma > 1$** , high-intensity (bright) pixels become darker.

Application Process

1. Read the input image.
2. Normalize pixel values to the range 0–1.
3. Apply the gamma transformation using **$\gamma > 1$** (e.g., $\gamma = 2.0$).
4. Convert the pixel values back to the 0–255 range.
5. Display the enhanced image.

Expected Outcome

- Excessive brightness is reduced.
- Bright regions become clearer.
- Hidden details become visible.
- Overall image visibility and contrast improve.

Justification

Gamma correction with $\gamma > 1$ is suitable because it decreases the intensity of overly bright pixels without significantly affecting darker regions. This produces a more balanced and visually pleasing image.

Result:

Input: Over-bright image → **Output:** Balanced brightness with improved visibility and clearer details.

2.A dark image lacks detail.

Describe and justify an appropriate grey level transformation to enhance visibility, explaining how it improves low-intensity regions with proper reasoning.

ANS:-Grey Level Transformation: Logarithmic (Log) Transformation

Concept:

Log transformation increases the brightness of dark pixels and reveals hidden details.

Formula:

$$s = c \log(1 + r)$$

where r = input pixel value, s = output pixel value, and c = constant.

Application:

1. Read the dark image.
2. Apply the log transformation.
3. Scale pixel values to 0–255.
4. Display the enhanced image.

Expected Outcome:

- Brightens dark regions.
- Reveals hidden details.

- Improves visibility and contrast.

Justification:

Log transformation expands low-intensity pixel values, making dark images clearer and easier to see. It is the best choice for enhancing images with poor brightness.

3. An image has poor contrast.

Explain a suitable histogram-based technique for contrast enhancement and illustrate how it improves image quality with logical reasoning.

ANS:-Histogram-Based Technique: Histogram Equalization

Concept:

Histogram Equalization improves image contrast by redistributing pixel intensity values over the entire gray-level range.

Application:

1. Read the input image.
2. Calculate the histogram.
3. Compute the cumulative distribution function (CDF).
4. Map old pixel values to new values using the CDF.
5. Display the enhanced image.

Expected Outcome:

- Improves overall contrast.
- Makes dark and bright regions more distinct.
- Reveals hidden details.
- Produces a clearer and sharper image.

Justification:

Histogram Equalization spreads pixel intensities uniformly, increasing the difference between light and dark areas. Therefore, it is the most suitable technique for enhancing low-contrast images.

4. A low-contrast image requires uniform intensity distribution.

Describe and justify the method you would apply, explaining the concept, working process, and its effect on intensity levels.

ANS:- A low-contrast image requires uniform intensity distribution.

Method: Histogram Equalization

Concept: Histogram Equalization redistributes pixel intensity values so they are spread more uniformly across the entire gray-level range, improving image contrast.

Working Process: Read the input image, calculate the histogram, compute the cumulative distribution function (CDF), map the old intensity values to new values using the CDF, and display the enhanced image.

Effect on Intensity Levels: Pixel intensities are distributed more evenly, increasing the difference between dark and bright regions. This improves contrast, reveals hidden details, and produces a clearer and more balanced image.

Justification: Histogram Equalization is the most suitable method because it creates a nearly uniform intensity distribution, enhancing the overall visibility and quality of low-contrast images.

5. An image shows uneven brightness across regions.

Explain and apply an appropriate enhancement technique to correct non-uniform illumination, providing a structured and accurate explanation.

ANS:- An image shows uneven brightness across regions.

Technique: Adaptive Histogram Equalization (AHE) / Contrast Limited Adaptive Histogram Equalization (CLAHE)

Concept: Adaptive Histogram Equalization enhances contrast by dividing the image into small regions and applying histogram equalization to each region separately. CLAHE further limits contrast to avoid noise amplification.

Working Process: Divide the image into small blocks, apply histogram equalization to each block (or CLAHE), combine the enhanced blocks smoothly, and display the corrected image.

Expected Outcome: Corrects non-uniform illumination, improves local contrast, enhances details in both dark and bright regions, and produces a uniformly illuminated image.

Justification: CLAHE is the most suitable technique because it enhances each region independently, effectively correcting uneven brightness while preventing excessive noise and over-enhancement.

6. A noisy image contains random intensity variations.

Explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.

ANS:- Smoothing Filter: Median Filter

Concept: A Median Filter removes random noise by replacing each pixel with the median value of its neighboring pixels while preserving image edges.

Working Process: Select a small neighborhood (e.g., 3×3), arrange the pixel values in ascending order, replace the center pixel with the median value, and repeat for all pixels.

Expected Outcome: Reduces random (salt-and-pepper) noise, preserves edges, and produces a smoother and clearer image.

Justification: The Median Filter is the most suitable smoothing filter because it effectively removes random noise without blurring important edges, resulting in better image quality.

7. An image requires noise reduction without significant loss of details.

Describe and apply an appropriate filtering approach, explaining how it balances noise removal and detail preservation.

ANS:- Filtering Approach: Bilateral Filter

Concept: A Bilateral Filter reduces noise while preserving edges by considering both the spatial distance and the intensity difference between neighboring pixels.

Working Process: For each pixel, calculate weights based on pixel distance and intensity similarity, compute the weighted average of neighboring pixels, and replace the original pixel with the filtered value.

Expected Outcome: Reduces noise, preserves edges and fine details, and produces a smooth, high-quality image.

Justification: The Bilateral Filter is suitable because it removes noise without blurring important edges, providing a good balance between noise reduction and detail preservation.

8. An image appears blurred after smoothing.

Explain and justify a suitable sharpening method to restore details, including its working principle and expected results.

ANS:- Sharpening Method: Laplacian Filter

Concept: The Laplacian Filter enhances edges by detecting sudden changes in pixel intensity, making the image appear sharper.

Working: Apply the Laplacian operator to detect edges and add the result to the original image to increase sharpness.

Result: Restores lost details, improves edge visibility, and produces a clearer and sharper image.

Justification: The Laplacian Filter is suitable because it enhances important image details without significantly changing the overall brightness.

9. Fine details in an image are not clearly visible.

Describe and justify the filtering technique used to enhance details, explaining its effect on high-frequency components.

ANS:- Filtering Technique: High-Pass Filter

Concept: A High-Pass Filter enhances fine details by emphasizing high-frequency components such as edges and textures.

Working: Apply the high-pass filter to highlight rapid intensity changes while suppressing smooth regions.

Result: Improves edge sharpness, enhances fine details, and makes the image clearer.

Justification: A High-Pass Filter is suitable because it strengthens high-frequency components, making small details and edges more visible without affecting the overall image structure.

10. A noisy image must be preprocessed before further analysis.

Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.

ANS:- Spatial Filtering Method: Median Filter

Concept: The Median Filter removes noise by replacing each pixel with the median value of its neighboring pixels while preserving edges.

Working: Select a neighborhood (e.g., 3×3), find the median value, replace the center pixel, and repeat for the entire image.

Result: Reduces noise, preserves edges, and produces a smoother and clearer image for further analysis.

Justification: The Median Filter is suitable because it effectively removes random noise while maintaining important image details, making it ideal for preprocessing.